🧬 Cell Biology – Assignment Questions

Section A: Short-Answer Questions

1. Define a **cell** and explain why it is considered the basic unit of life.
2. State two differences between **prokaryotic** and **eukaryotic** cells.
3. What is the function of the **nucleus** in a cell?
4. Name two organelles involved in **protein synthesis**.
5. What is the role of the **mitochondria**?
6. Define **osmosis**.
7. What is the difference between **diffusion** and **active transport**?
8. Name one structure found in plant cells but not in animal cells.
9. What is the function of the **Golgi apparatus**?
10. Define **cell membrane** and state one of its functions.

Section B: Long-Answer / Structured Questions

1. Describe the structure of the **plasma membrane** using the fluid mosaic model.
2. Explain the stages of the **cell cycle**, including mitosis and interphase.
3. Compare and contrast **mitosis** and **meiosis** in terms of purpose, process, and outcome.
4. Describe the structure and function of **ribosomes**, and explain why they are essential for cell survival.
5. Explain how **chloroplasts** enable plants to produce their own food.
6. Discuss the importance of **cell specialization** in multicellular organisms.
7. Describe the process of **endocytosis** and **exocytosis**, giving one example of each.
8. Explain how **enzymes** function in cellular metabolism and what factors affect their activity.

Section C: Application & Higher-Order Thinking

1. A plant cell is placed in a **hypertonic solution**. Predict what will happen and explain why.

2. A student observes a cell with no nucleus, circular DNA, and ribosomes. Identify the type of cell and justify your answer.
3. Explain why mitochondria are often called the **“powerhouses of the cell.”**
4. A mutation damages the lysosomes in a cell. Predict two consequences for the cell.
5. Explain how the structure of the **cell membrane** allows it to control what enters and leaves the cell.
6. Describe how **ATP** is produced during cellular respiration.
7. A scientist discovers a new organism with chloroplasts, a cell wall, and a large central vacuole. What type of organism is it likely to be? Explain your reasoning.